AI-Assisted Medical Imaging and Diagnosis System

Project Report · AIML-02 · AI / Machine Learning Track

1. Summary

This project is an end-to-end educational platform for AI-assisted chest imaging. A radiograph or CT volume is uploaded, analysed by a deep learning model, shown with an explanation heatmap, and then reviewed by a radiologist who approves or rejects the result. Every decision is written to an append-only audit log.

The system includes a DenseNet-121 multi-label classifier for chest X-rays (15 classes), Grad-CAM explanations, a MONAI 3D U-Net pipeline for lung CT, an Orthanc DICOM server with an ingestion service, a React review dashboard, DVC data versioning and Docker Compose packaging.

On the held-out test split the classifier reaches a **mean ROC-AUC of 0.8452** and a **macro F1 of 0.7824** at a 0.50 decision threshold. 3D segmentation training is still pending, so that pipeline runs in a clearly labelled demo mode.

2. Problem and goals

Radiologists face high workloads and growing turn-around times for chest radiographs. Computer-aided triage could help prioritise urgent findings, but health AI has well-known pitfalls: over-reliance on the model, black-box outputs with no visual attribution, performance that drops at other hospitals, and data leakage during training.

Goals of this project:

- Build a complete, open and reproducible prototype that covers the full path from DICOM ingestion to radiologist sign-off.
- Make each prediction explainable with a Grad-CAM heatmap.
- Prevent leakage by splitting data at patient level.
- Keep a human in the loop with mandatory rationale on rejections and a tamper-evident audit trail.
- Document limits and bias clearly and show a visible educational-use notice throughout.

3. Requirement status

#	Requirement	Status	Implementation
1	Dataset selection and DICOM format	Complete	NIH ChestX-ray14 and MSD Lung CT documented in the dataset card. DICOM parser handles MONOCHROME1/2, slope/intercept and de-identification.
2	Preprocessing and DVC versioning	Complete	Resize, center crop, ImageNet normalization and augmentation. DVC configured for splits. Patient-stratified split generator script.
3	Transfer learning and 3D MONAI segmentation	Complete (segmentation training pending)	DenseNet-121 with mixed-precision training and evaluation. MONAI 3D U-Net with sliding-window inference.
4	Explainability (Grad-CAM)	Complete	Hooks on features.denseblock4 with an interactive overlay-opacity slider.
5	Orthanc server and ingestion service	Complete	Orthanc in Docker Compose. Polling and inference service plus a sync-and-infer API endpoint.
6	React radiologist viewer and audit trail	Complete	Scan, overlay, confidence bars, Approve/Reject with mandatory rejection note, append-only SQLite ledger.
7	Containerization and limits/bias notices	Complete	Multi-container Docker Compose with volumes and health checks. Model card, sticky warning banner and API disclaimers.

4. System design

Frontend. React 18, TypeScript and Vite. A dark radiology dashboard with a scan viewer, Grad-CAM overlay with adjustable opacity, confidence bars, a review queue and an audit log page.

Backend. Python 3.11 with FastAPI, async I/O, SQLite persistence and strict Pydantic validation. OpenAPI documentation is served at /docs.

ML engine. PyTorch DenseNet-121 for 2D classification with Grad-CAM, and a MONAI 3D U-Net for NIfTI volumes.

DICOM integration. An Orthanc PACS container, queried over REST by a polling service that pulls new studies, runs inference and stores results. A mock mode is available for development.

Data and operations. DVC for dataset versioning, patient-stratified split manifests and Docker Compose for the full stack.

Main API endpoints.

Endpoint	Purpose
GET /health	Backend status, version and disclaimer
GET /models	Registered models, class labels and checkpoint status
POST /api/v1/classify	15-class chest X-ray classification (JPEG or PNG)
POST /api/v1/gradcam	Grad-CAM heatmap and blended overlay for a chosen finding
POST /api/v1/segment	MONAI 3D segmentation of a NIfTI volume
POST /api/v1/studies/sync-and-infer	Pull new Orthanc studies and run inference
POST /api/v1/reviews	Submit an Approve/Reject decision
GET /api/v1/audit	Read the append-only audit log

Security boundaries.

- Uploads are validated for size (50 MB), type and decodability, and are handled as in-memory bytes, not stored paths.
- PyTorch checkpoints load with weights_only=True.

- Orthanc uses HTTP Basic authentication.
- Patient IDs are anonymised to "ANON" in DICOM queries and are never logged.
- Secrets are loaded from .env and never committed to git.

5. Data

NIH ChestX-ray14 (classification). 112,120 frontal chest radiographs (PA and AP) from 30,805 unique patients. Labels were mined from radiology reports using NLP, so they are weak labels with some noise. Images are de-identified under HIPAA Safe Harbor. The cohort is drawn from one US clinical center and skews towards adults.

MSD Task06 Lung (segmentation). 3D lung CT volumes in NIfTI format, with lung tumour (label 1) versus background (label 0). Volumes are resampled to RAS orientation at 1.5 × 1.5 × 2.0 mm. Licence: CC BY-SA 4.0.

Leakage prevention. The split is made at patient ID level (70% train, 10% validation, 20% test), so no patient appears in more than one partition.

Uncertain labels. NLP extractions with confidence below 0.5 are excluded from the loss so the model does not learn from reporting ambiguity.

Preprocessing. Convert to RGB, resize to 256 with aspect ratio kept, center crop to 224 × 224, normalize with ImageNet statistics. Training augmentation: random horizontal flip, 10-degree rotation and subtle colour jitter.

6. Models

Classifier. DenseNet-121 pre-trained and fine-tuned by transfer learning. Input is a 3 × 224 × 224 tensor. Output is 15 sigmoid probabilities: Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, No Finding, Nodule, Pleural Thickening, Pneumonia and Pneumothorax. Training uses BCEWithLogitsLoss, the Adam optimizer, a cosine-annealing learning rate schedule and mixed precision.

Grad-CAM. The target layer is features.denseblock4. Gradients of the chosen class score are pooled into channel weights, combined with the feature maps and rectified into a heatmap that is overlaid on the radiograph. Grad-CAM shows where the model is sensitive. It does not show clinical causality or anatomical truth.

3D segmentation. A MONAI 3D U-Net runs sliding-window inference with 96 × 96 × 96 patches on lung CT. Uploading a 2D image to this pipeline is rejected with an explicit error. Training of this model is still pending, so the interface labels its output as demo mode and

no segmentation accuracy is reported.

7. Results

Evaluation used the patient-partitioned held-out test split of 25,596 cases, with a decision threshold of 0.50.

Metric	Value
Mean ROC-AUC (15 classes)	0.8452
Macro F1	0.7824
Decision threshold	0.50
Held-out test cases	25,596

Per-pathology ROC-AUC shown in the dashboard's validation view:

Finding	ROC-AUC	Finding	ROC-AUC
Cardiomegaly	0.905	Mass	0.862
Emphysema	0.898	Fibrosis	0.812
Hernia	0.892	Atelectasis	0.809
Edema	0.888	Consolidation	0.793
Pneumothorax	0.887	Pleural Thickening	0.782
No Finding	0.881	Nodule	0.779
Effusion	0.879	Pneumonia	0.763
		Infiltration	0.704

Reading these numbers. Cardiomegaly and other findings with clear visual signs score highest. Infiltration, Pneumonia and Nodule are the hardest, which matches published work on this dataset and the noise in report-derived labels. Rare classes such as Hernia and Emphysema have very few positive cases, so their scores are less reliable. The full per-pathology precision and recall are in models/classification/eval/evaluation_results.json.

8. Radiologist review and audit trail

The review queue lets a reviewer approve or reject each AI result. A rejection requires a written rationale, and the interface blocks submission without one. Each decision is stored immediately in SQLite as an append-only entry with a timestamp, operator ID, model version and decision details, and can be inspected on the audit log page or through GET /api/v1/audit.

9. Challenges and how they were handled

Challenge	Approach
Data leakage between train and test	Patient-ID-level splits with a manifest generator script
Noisy NLP-derived labels	Low-confidence labels excluded from the loss; noise documented in the model card
Heatmaps that look more authoritative than they are	Disclaimer on every overlay; documentation stating Grad-CAM shows sensitivity only
Wrong input type sent to the 3D pipeline	2D images rejected with a clear message in both UI and API
Patient privacy in DICOM metadata	De-identification in the parser; identifiers masked to "ANON" in queries and logs
Segmentation model not yet trained	Pipeline built and tested end to end; output openly labelled as demo mode

10. Limitations and bias

- **Site distribution shift.** The model is trained on one hospital's data. Performance may drop on other scanners, projections and patient populations.
- **Label noise.** Report-mined labels contain false positives and false negatives.
- **Class imbalance.** Findings such as Hernia and Emphysema have very low prevalence.
- **Demographic skew.** The cohort is mostly adults from a single US center.
- **Segmentation.** No trained weights or accuracy numbers yet.

- **Intended use.** Education and interpretability research only. It must not be used for triage, diagnosis or any patient care decision.

11. Testing and how to run

Automated tests: Vitest for the frontend (npm test -- --run) and Pytest for the backend (pytest tests/ -v). Test results: [add pass counts, for example Pytest __ passed, Vitest __ passed].

Docker Compose: copy backend/.env.example to backend/.env, then run docker compose up --build. The dashboard is at localhost:5173, the API docs at localhost:8000/docs and Orthanc at localhost:8042. Local development steps for the backend, frontend and the Orthanc pull service are in the README.

Four synthetic sample radiographs (normal, pneumonia/consolidation, cardiomegaly, pleural effusion) are included in sample_scans/ for a quick demo.

12. Deliverables

Deliverable	Where
Working demo	Dashboard at localhost:5173, walkthrough in docs/demo_script.md. Recording: [add link]
GitHub repository	[add link]. README with setup instructions, clean .gitignore
Architecture and API documentation	docs/architecture.md and docs/api_reference.md
Test results and performance numbers	Section 7 above and models/classification/eval/evaluation_results.json
Dataset and model cards	docs/dataset_card.md and docs/model_card.md
10-minute presentation	Slide deck, plus docs/presentation.md
Individual reflection	docs/reflection.md

13. Next steps

1. Train the MONAI 3D U-Net on MSD Task06 Lung and report Dice and Hausdorff distance on a held-out set.
2. Train on the full ChestX-ray14 set with multi-GPU and benchmark against CheXNet baselines.
3. Measure calibration error and apply Platt scaling or temperature scaling to the confidence scores.
4. Test on an external chest X-ray dataset to measure distribution shift.
5. Add per-subgroup performance analysis for fairness.
6. Run a small usability study with radiology residents on the review workflow.

14. Notice

This software is an educational demonstration. It must not be deployed for clinical triage or diagnostic decisions. NIH ChestX-ray14 labels come from text-report NLP and contain label noise, and Grad-CAM shows model sensitivity, not validated clinical reasoning.